

WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University

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EDUCATION

Nanyang Technological University, Singapore

Expected 2028

Bachelor of Science in Physics

CGPA: 4.38/5.00

Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

From the Quantum Particle to Quantum Field Theories: A Mathematical Tour

August 2026-Present

URECA Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Developing a mathematically structured route from single-particle quantum mechanics to nonrelativistic and relativistic quantum field theories.
- Constructing a dictionary between one-particle and second-quantized formulations, covering Fock spaces, creation and annihilation operators, operator-valued fields, and selected foundational aspects of Yang-Mills theory.

Memory Effects in Stochastic Thermodynamics

May 2026-Present

Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the framework to the full non-Markovian system, examining fluctuation-dissipation consistency, stationary behavior, and entropy production.

RESEARCH INTERESTS

Undergraduate focus: Non-Markovian stochastic dynamics; stochastic thermodynamics and nonequilibrium statistical mechanics; mathematical foundations of quantum mechanics and quantum field theory; foundations of Yang-Mills theory.

Future directions: Symmetry and representation theory; holography and AdS/CFT; quantum information theory.

SELECTED ACADEMIC WRITING

Analytical Mechanics

Available on website

Structured notes on Lagrangian and Hamiltonian mechanics.

Classical Field Theory

In preparation

Quantum Mechanics

In preparation

Mathematical Methods for Physics

In preparation

TECHNICAL PROJECTS AND LEADERSHIP

Illuminatio Physica

2026-Present

Creator and Developer · illuminatio.techecho.org

- Designed and developed an interactive 3D knowledge graph for exploring physics concepts, semantic relationships, prerequisites, and learning paths.

Tech Echo Collective

2026-Present

Founder and Developer · techecho.org

- Founded and developed a multilingual open-source platform for technical projects and long-form discussion, with GitHub-based identity, project attribution, and contributor history.